LAB 3
TOTAL INTERNAL REFLECTION

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OBJECTIVE AND EXPERIMENT

- Rotating a trapezoidal acrylic prism to determine the index of refraction and the critical angle between air and acrylic

NORMAL
COMPONENT
COMPONENT
RAY TABLE
DEGREE SCALE
SLITS
5 3 1
RAY
COLOR
PATTERN
LIGHT
SOURCE
OS-8517A
TO REPLACE BULB SEE
INSTRUCTION MANUAL.
BULB: 12V 10W G-4
DO NOT TOUCH BULB
WITH FINGERS.
DISCONNECT POWER
BEFORE CHANGING BULB.

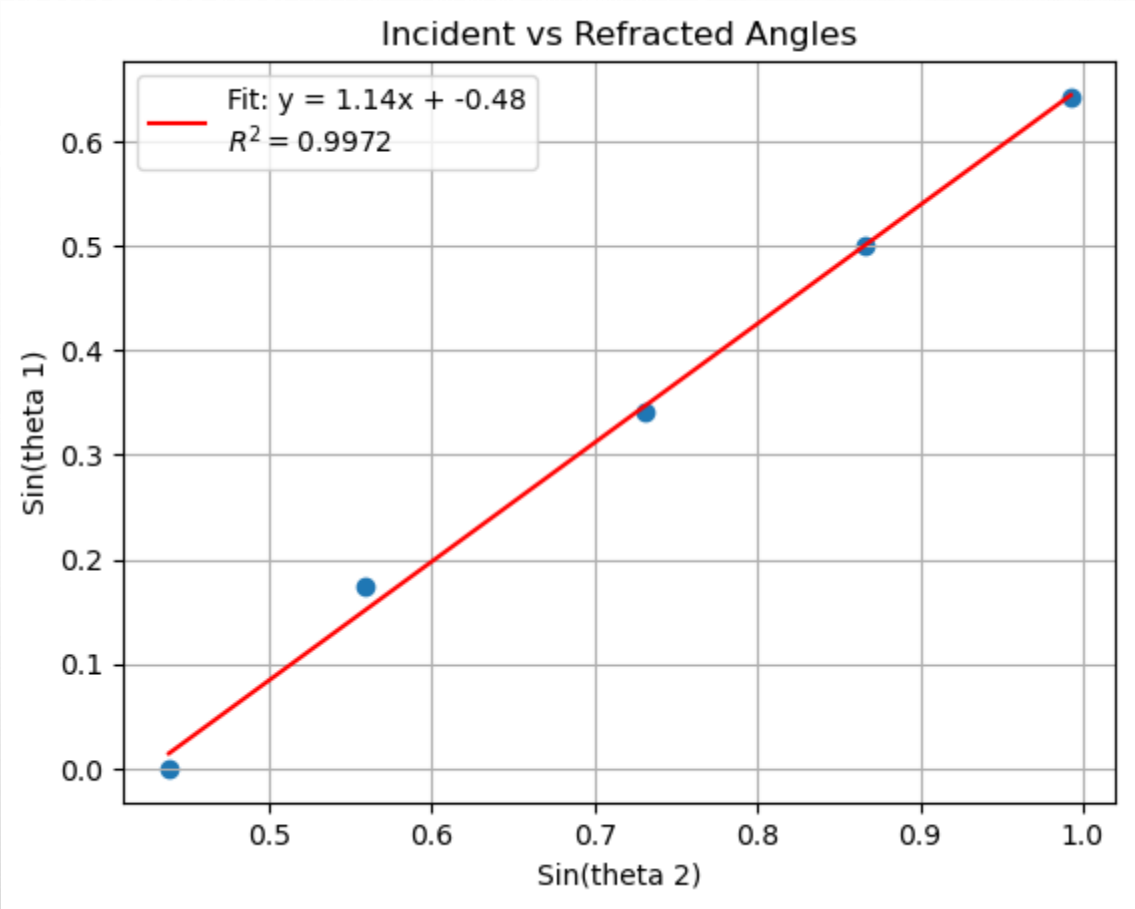
SETUP

COMPONENT
COMPONENT
NORMAL
LIGHT SOURCE
PATTERN
COLOR
RAY
SLITS
5 3 1

SETUP

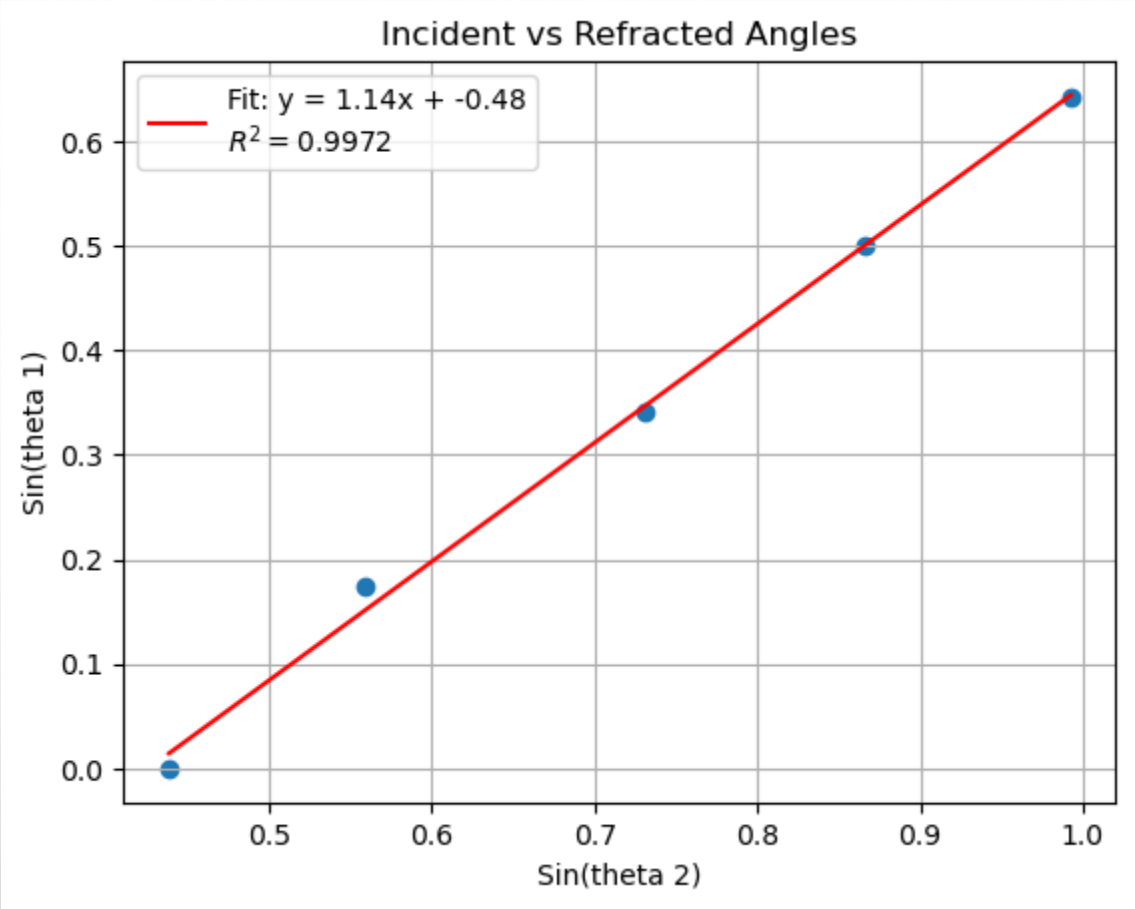
GRAPH

- θ_1 is the refracted angle
- θ_2 is the angle of incidence
- $m = \frac{\sin(\theta_1)}{\sin(\theta_2)} = \frac{n_2}{n_1}$
- The calculated index of refraction of the acrylic is 1.14
- Then the calculated critical angle is $\sin^{-1}(1/m) = 61.2°$



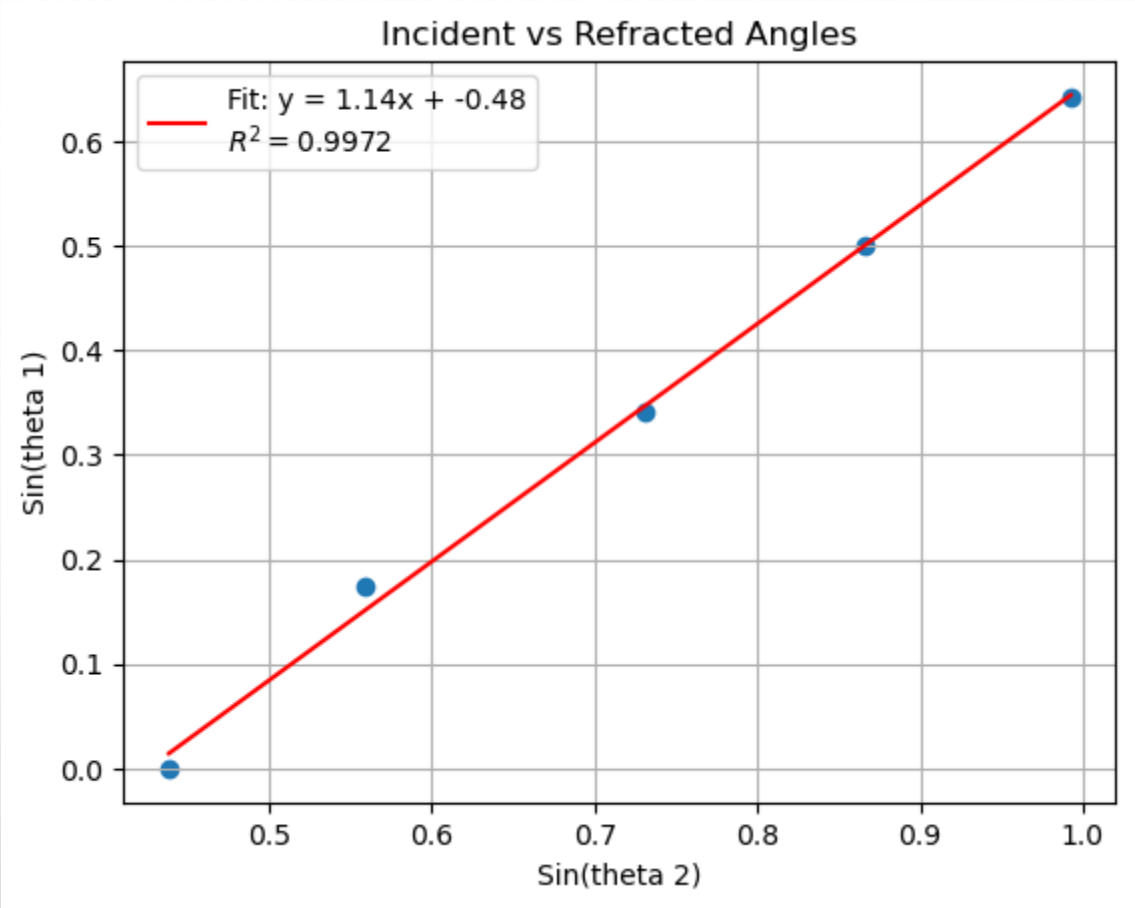
ERROR ANALYSIS

- The theoretical of the index of refraction is n = 1.5
- The theoretical critical angle of acrylic is 42.1 degrees
- We get a percent error of 25% with the index of refraction
- And a 43% error in the critical angle



ERROR ANALYSIS

- Uncertainty in the measurement was $\pm 1°$
- Lack of data near the critical angle
  - We only have 5 points, and they stop at 40°,
  - Without data points between 40° and the actual critical angle (where the light hits 90°), the linear fit is guessing the slope based on the flatter part of the curve.



CONCLUSION

- The index of refraction for the acrylic prism was determined using the slope of the linear fit ($m = 1.14$ a 25% error). Based on this, the critical angle between air and acrylic was calculated to be $61.2°$ with a 43% error
- Rotating the prism allowed for the successful observation of total internal reflection and provided necessary data to apply Snell's Law
- While the linear fit showed high correlation, the calculated critical angle is likely an overestimate due to a lack of data points beyond 40 degrees